EUROPEAN URBAN INITIATIVE

UNIVERSITÀ DELLA CALABRIA

TECHNICAL REPORT
CASTROVILLARI AND CASSANO ALL'IONIO. WILDFIRE RISK ANALYSIS

Multi-factor analysis of wildfire risk propensity, based on potential spread simulation and examination of the main contributing factors: vegetation, terrain morphology, water resources and historical ignition dynamics.

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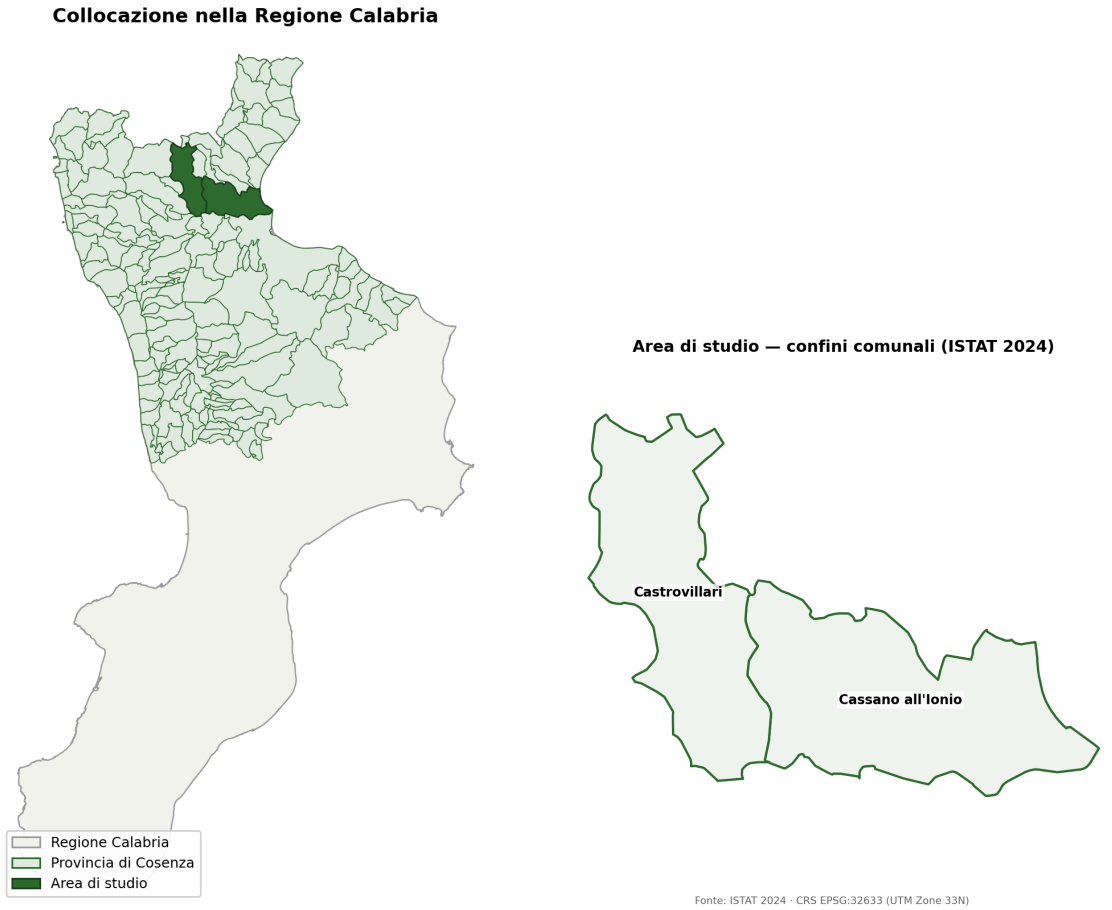
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1 General Overview

The municipalities of Castrovillari and Cassano all'Ionio are located in the province of Cosenza, on the northern slope of Calabria, on the edge of the Pollino National Park. The territory is characterised by a complex orography, from the inland hills to the Sibari plain towards the Ionian coast, with extensive forest cover that, combined with the summer climate, exposes the area to a significant wildfire risk.



(a) *Location within the Calabria Region.* **(b)** *Study area - municipal boundaries (ISTAT 2024).*

Figure 1: *Geographic overview of the study area.*

Table 1 summarises the main morphometric characteristics of the two municipalities.

Table 1: *Summary territorial data (source: ISTAT 2024, DEM 20 m, CRS EPSG:32633).*

Comune	Superficie (ha)	Quota min (m)	Quota max (m)	Quota media (m)
Cassano all'Ionio	14,017	0	661	90
Castrovillari	11,837	0	2262	580

Methodological Notes

The municipalities of Castrovillari and Cassano all'Ionio form a study area representative of the wildfire risk dynamics typical of northern Calabria. Their position on the edge of the Pollino National Park, combined with marked altitudinal variation - from 0 to over 2 200 m a.s.l. in Castrovillari, compared to 661 m in Cassano all'Ionio - defines deeply different territorial contexts, both from a vegetation and morphological standpoint. The total area of approximately 25 854 ha, entirely within the province of Cosenza, is characterised by a landscape mosaic ranging from coastal areas and the Sibari plain to the Apennine hinterland. This structural heterogeneity makes a multi-factor analysis necessary — one that goes beyond vegetation cover alone and integrates morphology, water resources and historical ignition dynamics to produce a spatially differentiated risk assessment.

2 Vegetation and Protected Areas

2.1 ISPRA Carta Natura - Vegetation Overview

The ISPRA Carta Natura, produced according to the EUNIS (*European Nature Information System*) classification, provides a complete picture of land cover and habitats in the study area. Forests and areas of natural vegetation represent the primary driver of potential fire spread in the area.

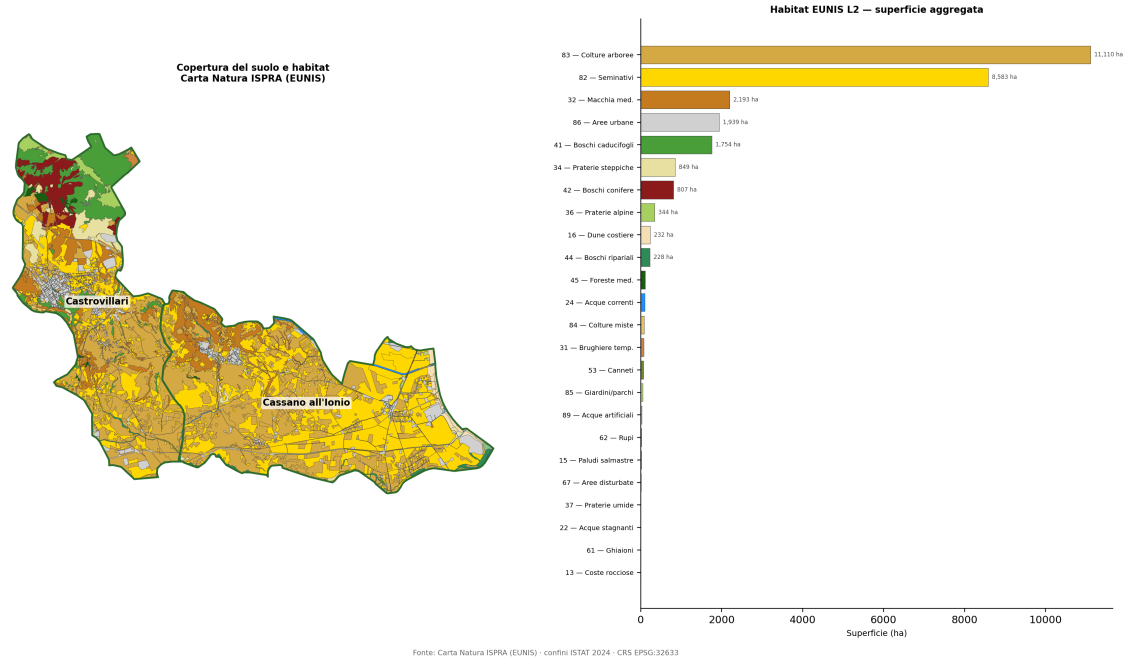


Figure 2: *Land cover and habitats (Carta Natura ISPRA, EUNIS classification) with area distribution by L2 habitat type. Source: Carta Natura ISPRA — confini ISTAT 2024 — CRS EPSG:32633.*

Table 2 lists the fifteen habitat types with the largest areal extent, ranked in descending order of surface area.

Table 2: *EUNIS L2 habitats - aggregated area in the study area (source: Carta Natura ISPRA).*

Codice EUNIS	Denominazione	Superficie (ha)
83	Colture arboree	11,110
82	Seminativi	8,583
32	Macchia med.	2,193
86	Aree urbane	1,939
41	Boschi caducifogli	1,754
34	Praterie steppiche	849
42	Boschi conifere	807
36	Praterie alpine	344
16	Dune costiere	232
44	Boschi ripariali	228
45	Foreste med.	110
24	Acque correnti	101
84	Colture miste	80
31	Brughiere temp.	79
53	Canneti	67

2.2 Rete Natura 2000 — ZSC/SIC e ZPS

The thematic raster classifies the territory according to the degree of overlap with Rete Natura 2000 sites, following EU legislation:

- **ZSC** (Zona Speciale di Conservazione, ex SIC — Sito di Importanza Comunitaria): habitat protection sites for natural habitats and wild flora and fauna, established under Directive 92/43/EEC; «Habitat»;
- **ZPS** (Zona di Protezione Speciale): wild bird protection sites, established under Directive 2009/147/EC; «Uccelli»;
- **ZSC + ZPS**: areas of overlap between both types of protected site.

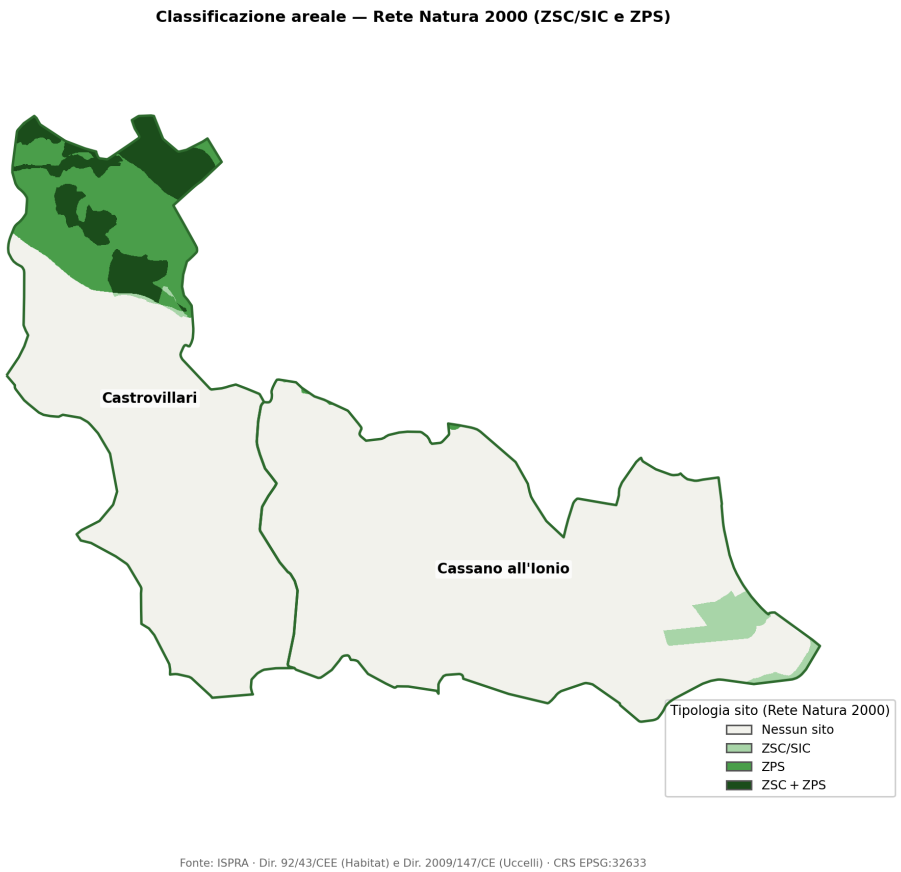


Figure 3: *Areal classification with respect to Rete Natura 2000 sites. Source: ISPRA — Dir. 92/43/CEE e Dir. 2009/147/CE — CRS EPSG:32633.*

Table 3: *Share of municipal area by Rete Natura 2000 site type. ZSC: Zona Speciale di Conservazione (ex SIC, Dir. 92/43/CEE); ZPS: Zona di Protezione Speciale (Dir. 2009/147/CE).*

Comune	Nessun sito	ZSC/SIC	ZPS	ZSC+ZPS
Cassano all'Ionio	96.5%	3.4%	0.1%	0.0%
Castrovillari	71.6%	0.3%	17.5%	10.6%

ZSC: Zona Speciale di Conservazione (Dir. 92/43/CEE Habitat); ZPS: Zona di Protezione Speciale (Dir. 2009/147/CE Uccelli).

Integrated Analysis

The territory of the two municipalities presents a landscape mosaic of considerable complexity. The large agricultural expanses of the plain - tree crops and arable land dominating the valley floor and coastal strip of Cassano all'Ionio - gradually give way, moving uphill towards the Castrovillari hinterland, to the natural and semi-natural formations typical of the Calabrian Apennines: Mediterranean scrub, deciduous woodland, steppe grasslands and, at higher elevations, the forests of the Pollino National Park. It is precisely this transition zone, where continuous vegetation meets significant slopes and increasingly dry summer conditions, that presents the greatest propensity for fire spread.

In terms of ecological protection, Castrovillari has the greatest overlap with the Natura 2000 network: over a quarter of its area falls within SAC/SPA sites, concentrated in the northern portion corresponding to the Pollino massif. Cassano all'Ionio is almost entirely outside the protection network, with small protected nuclei along the coastal strip. The presence of protected habitats does not reduce fire risk, but imposes specific constraints on intervention methods, making prevention the primary operational lever over active suppression.

3 Water Resources for Active Firefighting

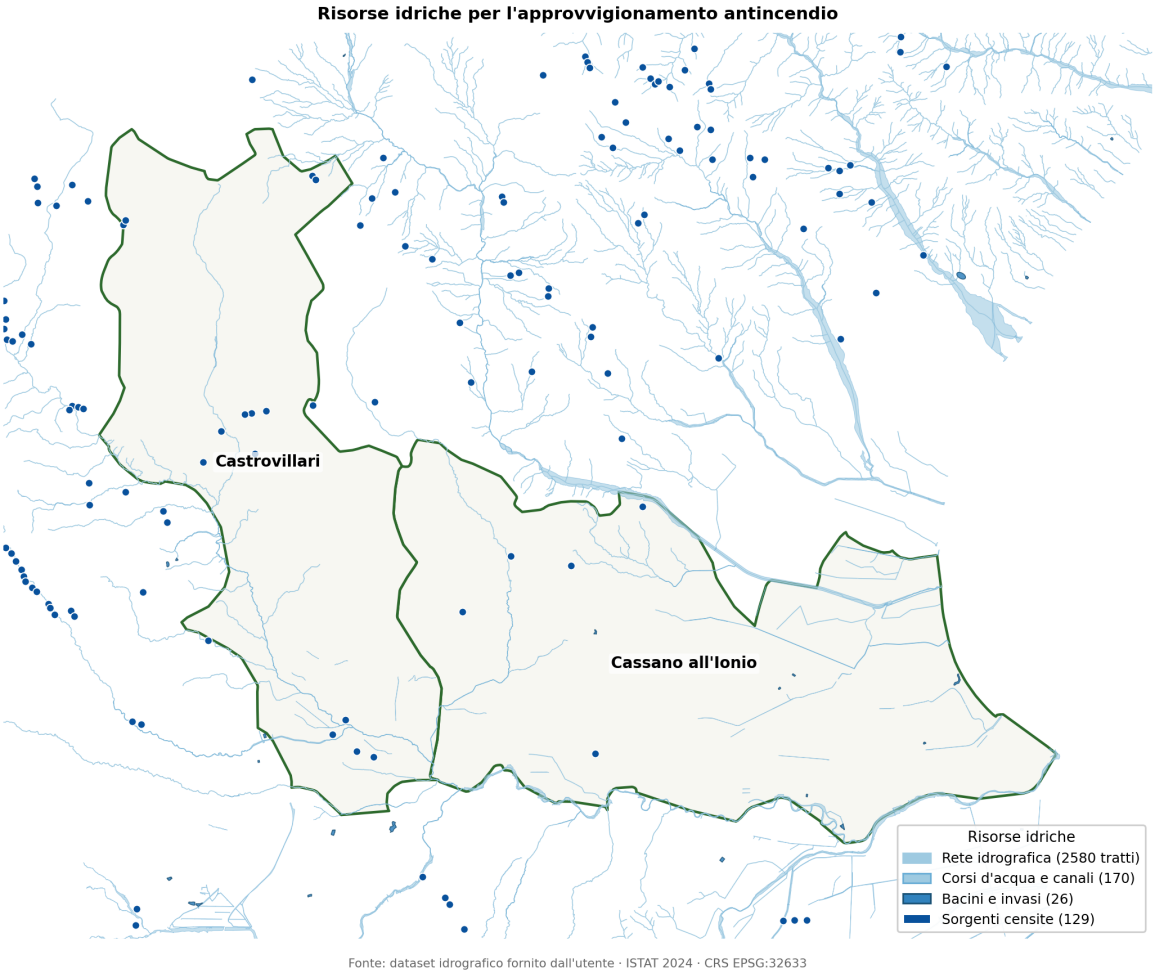


Figure 4: *Water resources available for firefighting supply: hydrographic network, watercourses and canals, reservoirs and basins, registered springs. Source: rilievo idrografico — ISTAT 2024 — CRS EPSG:32633.*

To complete the territorial picture, water resources are distributed across both municipalities through a dense hydrographic network supplemented by watercourses, reservoirs, basins and springs. The greatest concentration of supply points is found in the foothill belt of Castrovillari, while the inland areas of Cassano all'Ionio, despite being crossed by significant watercourses, have a relatively sparser distribution of abstraction points.

The effective availability of these resources during the summer months is, however, conditioned by the Mediterranean hydrological regime, which requires a detailed verification of the accessibility and residual flow of individual water points for firefighting planning purposes.

4 Terrain Morphology

4.1 Slope Classes

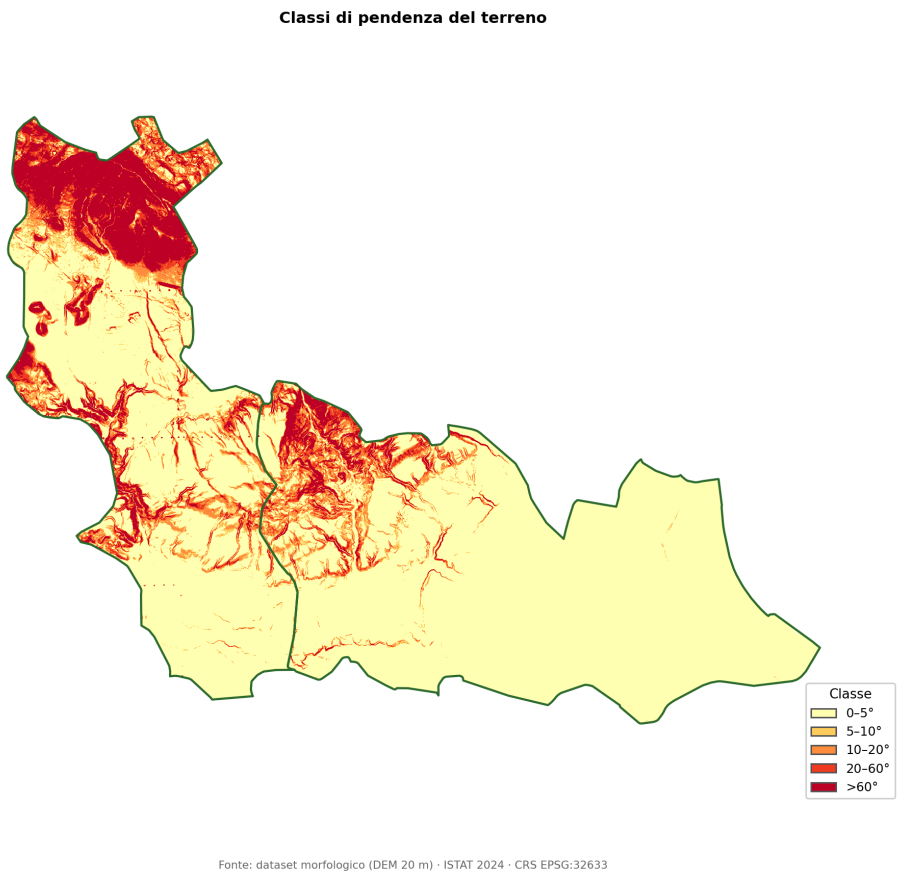


Figure 5: *Terrain slope classes derived from the 20 m DEM. Source: dataset morfologico — ISTAT 2024 — CRS EPSG:32633.*

Table 4: *Distribution of slope classes per municipality (% of total municipal area).*

Comune	0–5°	5–10°	10–20°	20–60°	>60°
Cassano all'Ionio	85.2%	2.8%	5.2%	3.7%	3.1%
Castrovillari	58.6%	5.5%	9.4%	7.9%	18.5%

4.2 Aspect Classes

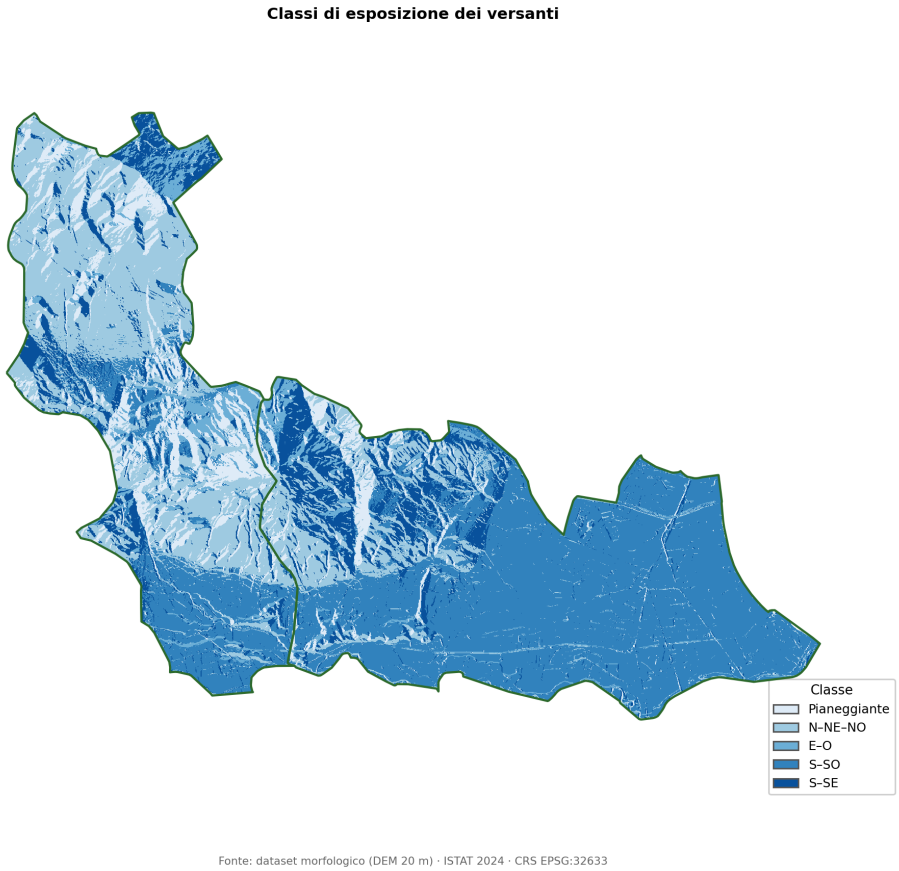


Figure 6: *Slope aspect classes derived from the 20 m DEM. Source: dataset morfologico — ISTAT 2024 — CRS EPSG:32633.*

Table 5: *Distribution of aspect classes per municipality (% of total municipal area).*

Comune	Pianeggiante	N–NE–NO	E–O	S–SO	S–SE
Cassano all'Ionio	5.2%	13.3%	4.4%	64.0%	13.1%
Castrovillari	17.5%	43.5%	7.9%	18.7%	12.4%

Morphological Analysis

The study area displays two deeply contrasting morphological characters, reflecting the transition between the coastal plain system of the Sibari plain and the Apennine ridges of the Pollino. Cassano all'Ionio develops predominantly on flat or gently sloping terrain, with mean elevations around 90 m a.s.l. This configuration favours a degree of homogeneity in risk distribution, but exposes large surfaces to the summer aridity typical of the Ionian coastal strip.

Castrovillari instead presents a far more varied altitudinal profile, with elevations ranging from the valley floor to over 2 200 m a.s.l. and markedly steep slopes, especially

in the montane portion. In this context, slope becomes an amplifier of fire behaviour: on steep terrain the fire front advances more rapidly uphill due to fuel preheating, reducing intervention windows. Aspect introduces a further variable: areas with greatest solar exposure develop more intense vegetation water stress, creating conditions of high flammability from the early weeks of the fire season.

The combination of slope, aspect and elevation defines a spatial hierarchy of morphological risk that does not follow administrative boundaries, but distributes itself along altitudinal transition zones where terrain complexity overlaps with continuous fuel loads.

5 Historical Ignition Analysis

In broad terms, an ignition is the event or process that initiates combustion - that is, the combination of three elements sufficient to start a self-sustaining oxidation reaction: fuel, oxidiser (oxygen) and heat. This is the well-known fire triangle.

Wildfire ignition:
In the context of forestry and civil protection, an ignition is the moment and location where fire first starts in vegetation. It may be caused by:
`Natural causes` — lightning strike, rarely spontaneous combustion under extreme conditions.
`Human causes` — which account for the overwhelming majority of cases in Italy, subdivided into:

- Accidental: sparks from agricultural machinery, cigarette butts, unattended fires, pruning residues burned on windy days.
- Deliberate: intentional arson, unfortunately the dominant cause in Calabria.
- Negligent: careless behaviour.

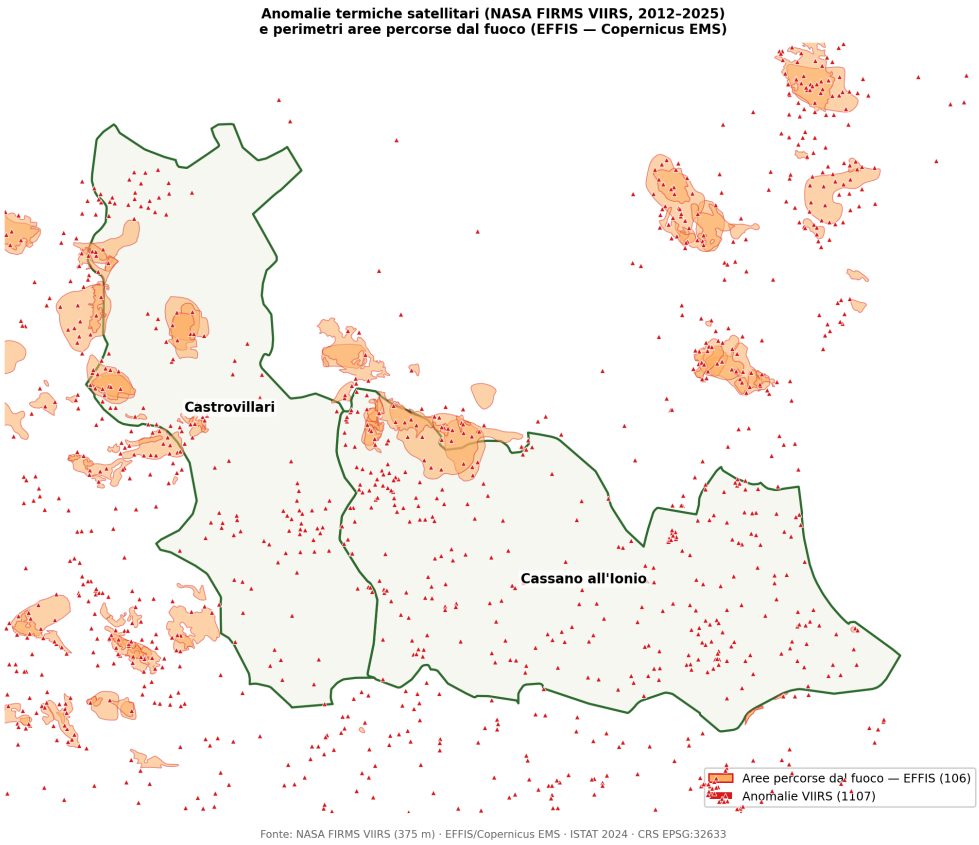


Figure 7: *Satellite thermal anomalies (NASA FIRMS — VIIRS, 375 m, 2012–2025) and burned area perimeters (EFFIS — Copernicus Emergency Management Service). Fonte: NASA FIRMS / EFFIS — ISTAT 2024 — CRS EPSG:32633.*

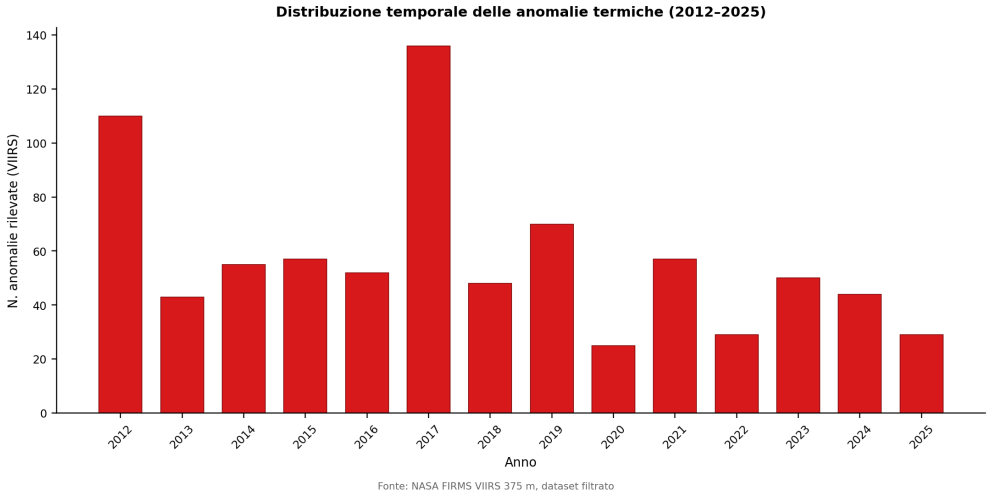


Figure 8: *Annual temporal distribution of thermal anomalies detected in the study area (NASA FIRMS VIIRS, 2012-2025).*

The historical ignition analysis, based on over a decade of satellite observations, returns a picture of the territory consistent with the simulated risk maps, confirming the validity of the predictive model adopted. The over 1 150 thermal anomaly points recorded between 2012 and 2025 are distributed non-randomly: the greatest concentration is found along the boundary zone between the two municipalities and in the western portion of Castrovillari. The year 2017 stands out as the peak, with over 200 anomalies recorded, consistent with the exceptionally intense fire season that affected the entire peninsula. In more recent years the number of recorded events tends to decrease, but this trend should be interpreted with caution: the structural risk conditions - vegetation, morphology, climate - have not changed significantly.

6 Tourist Exposure and Vulnerability

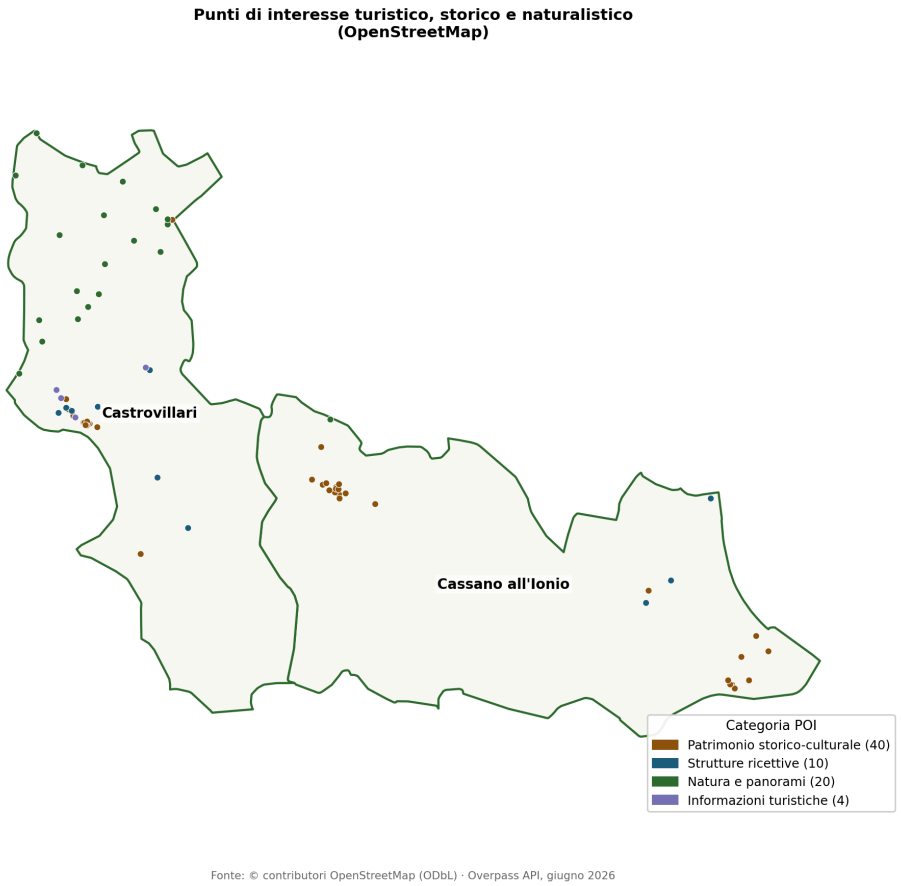


Figure 9: *Tourist, historical and naturalistic points of interest (OpenStreetMap, June 2026). Source: © OpenStreetMap contributors (ODbL).*

Table 6: *Tourist points of interest by municipality and category (source: OpenStreetMap, Overpass API, June 2026)*

Comune	Storico-cult.	Strutt. ricettive	Natura e pan.	Info turistiche	Totale
Cassano all'Ionio	25	3	1	0	29
Castrovillari	15	7	19	4	45
textbfTotale area	40	10	20	4	74

The analysed territory hosts 74 tourist, historical and naturalistic points of interest, with an asymmetric distribution between the two municipalities: Castrovillari concentrates the majority of naturalistic and scenic sites (19 out of 20), consistent with its proximity to the Pollino National Park, while Cassano all'Ionio has a predominantly historical and cultural character (25 POIs out of 29), attributable to the archaeological richness of the Sibari plain.

The presence of accommodation facilities - more numerous in Castrovillari (7 versus 3) - indicates tourist flows concentrated precisely in the areas of greatest morphological

and vegetation complexity, where fire risk is most pronounced. During the summer season, the overlap between peak tourist pressure and maximum ignition probability requires integrated planning that considers evacuation routes, early warning capacity along hiking trails and training of accommodation staff.

7 Wildfire Risk Maps

The risk maps are derived from the raster `copia_rischio.tif` and classify the territory into five operational classes (Table 7), aligned with the QGIS style file `rischio.qml`.

Table 7: *Wildfire risk class legend (QGIS classification `rischio.qml`, DISCRETE scheme).*

Valore	Classe	Colore	Soglia
0	Nodata		Assenza di dati
1	Low		$\leq 1{,}0$
2	Low-Medium		$1{,}0 - 2{,}0$
3	Medium		$2{,}0 - 3{,}0$
4	Medium-High		$3{,}0 - 4{,}0$
5	High		$> 4{,}0$

7.1 Area Overview

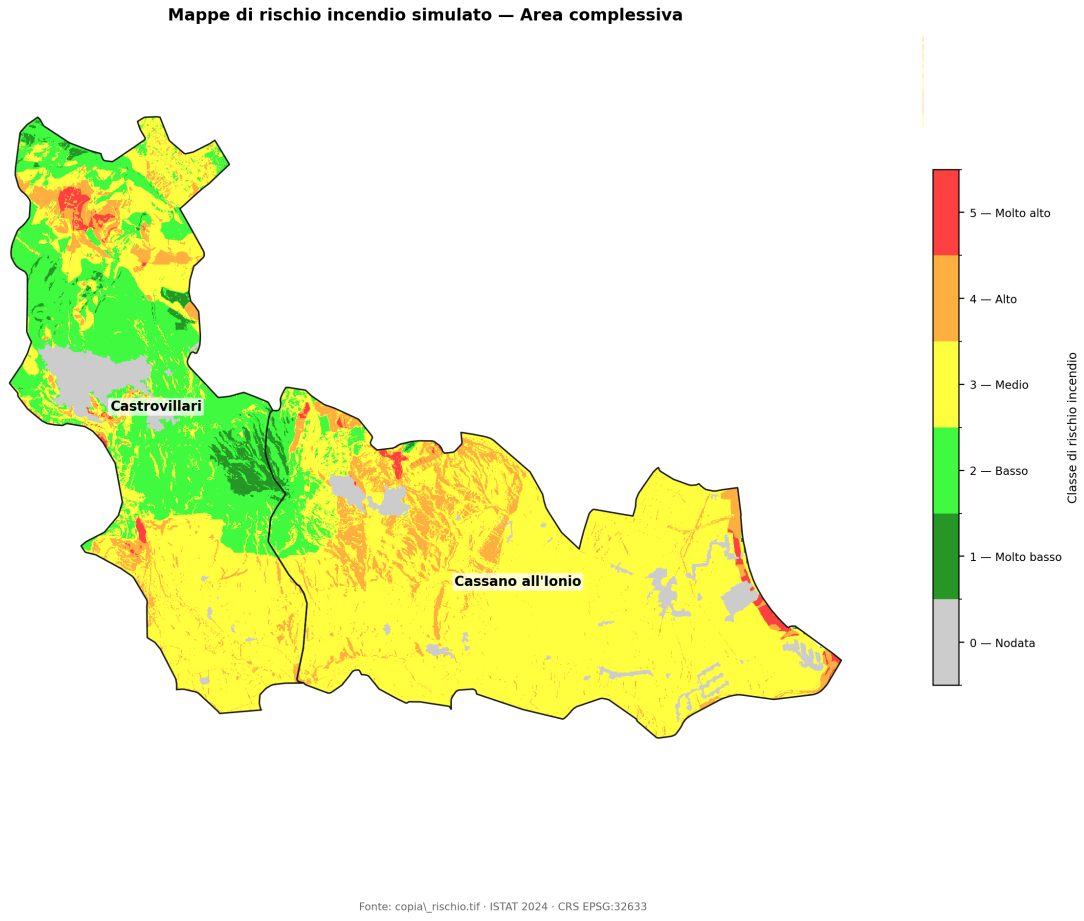
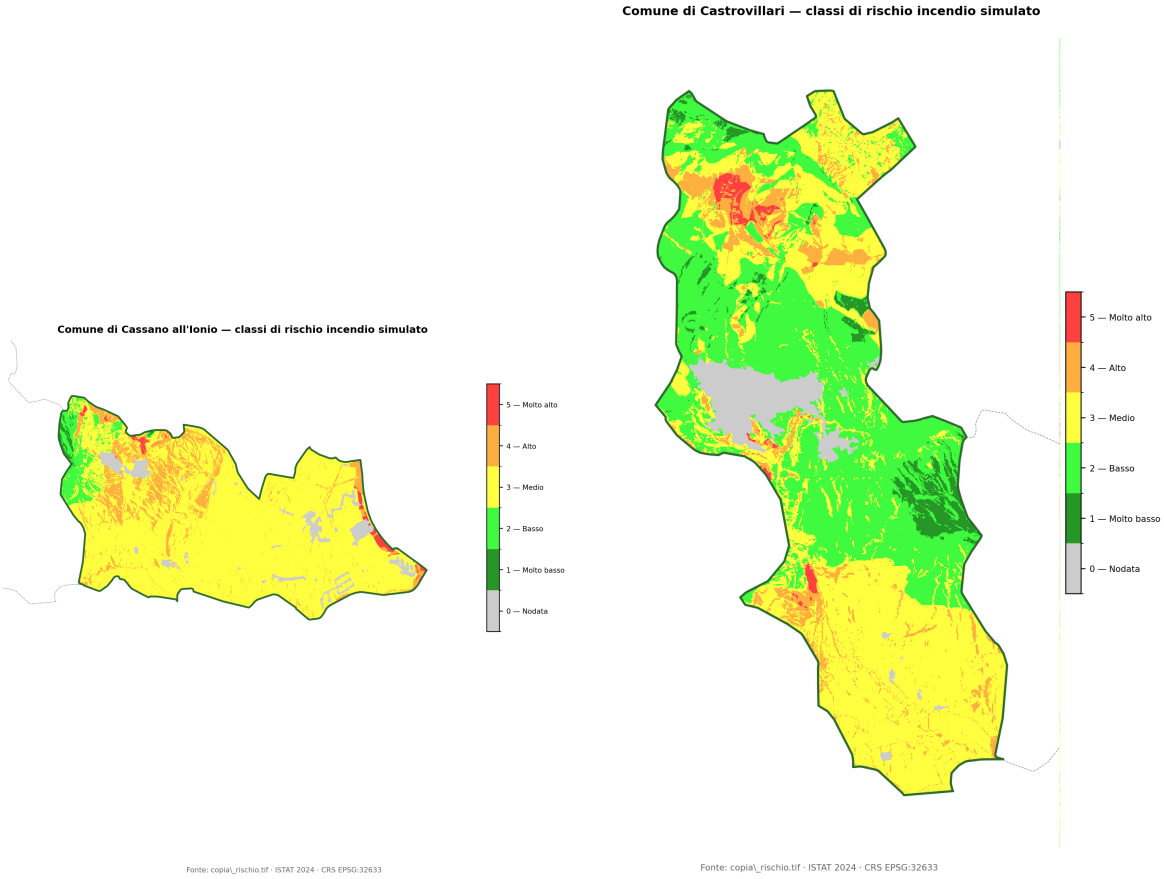


Figure 10: *Simulated wildfire risk map - complete area overview. Source: `copia_rischio.tif` - ISTAT 2024 — CRS EPSG:32633.*

7.2 Detail by Municipality



(a) *Municipality of Cassano all'Ionio.* **(b)** *Municipality of Castrovillari.*

Figure 11: *Wildfire risk maps — detail by municipality. Source:* `copia_rischio.tif` *— ISTAT 2024 — CRS EPSG:32633..*

7.3 Statistical Risk Distribution

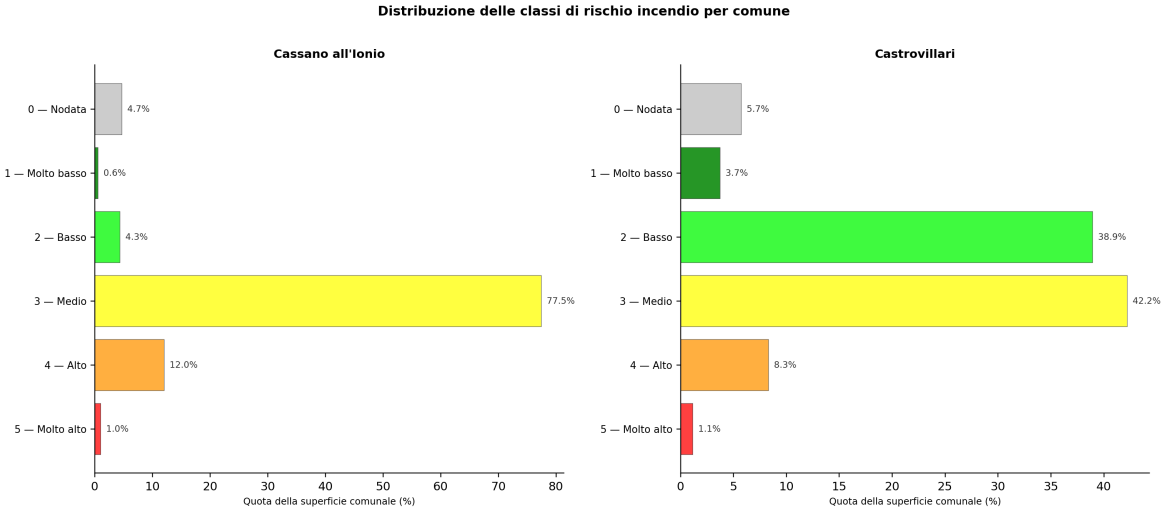


Figure 12: *Percentage distribution of wildfire risk classes per municipality (share of total municipal area).*

Table 8: *Percentage distribution of wildfire risk classes per municipality (% of total municipal area).*

Comune	Nodata	Molto basso	Basso	Medio	Alto	Molto alto
Cassano all'Ionio	4.7%	0.6%	4.3%	77.5%	12.0%	1.0%
Castrovillari	5.7%	3.7%	38.9%	42.2%	8.3%	1.1%

Risk Analysis

From a simulated risk perspective, the two municipalities show distinct but equally noteworthy profiles. Cassano all'Ionio, despite its predominantly flat terrain, records a higher mean risk, explained by the widespread presence of crops and Mediterranean scrub subject to strong summer water stress and by the historical recurrence of ignitions along the agricultural strip.

Castrovillari instead displays a more polarised risk distribution: large portions of the municipality fall into low or medium classes, corresponding to higher elevations where montane vegetation and greater relative humidity reduce combustion propensity; mid-elevation areas - where slope, aspect and shrub cover combine with proximity to historical ignition points - concentrate the most critical risk values. In both municipalities, localised nuclei of very high class are identified that require priority attention in the planning of prevention and surveillance activities.

8 Summary Table

Table 9: *Risk statistics summary by municipality. Mean R.: mean risk raster value; P90 R.: 90th percentile; Max R.: maximum value.*

Comune	Sup. (ha)	Quota min	Quota max	Quota media	R. medio	R. P90	R. max
Cassano all'Ionio	14017	0 m	661 m	90 m	2.94	4.00	5.00
Castrovillari	11837	0 m	2262 m	580 m	2.47	3.00	5.00

Data References and Sources

Wildfire risk `copia_rischio.tif` — discrete classification 0–5, CRS EPSG:32633, resolution 20 m.

Admin. boundaries ISTAT — Municipal administrative boundaries 2024
https://www.istat.it

Natura Map ISPRA — Carta della Natura, EUNIS classification
https://www.isprambiente.gov.it

Natura 2000 ISPRA — SAC/SCI and SPA dataset, Dir. 92/43/EEC and Dir. 2009/147/EC.

Morphology 20 m DEM — slope and aspect classes, thematic dataset provided by the user.

Hydrography Hydrographic survey (springs, basins, watercourses) provided by the user.

VIIRS ignitions NASA FIRMS — VIIRS Active Fire Detection, 375 m
https://firms.modaps.eosdis.nasa.gov

Burned areas EFFIS — Copernicus Emergency Management Service
https://effis.jrc.ec.europa.eu

Tourist POIs © OpenStreetMap contributors (ODbL), Overpass API, June 2026
https://www.openstreetmap.org

Reference CRS: EPSG:32633 — WGS84 / UTM Zone 33N. All areas in hectares (ha). Processing: June 2026.